

If AI uses so much water, is there a way to use dirty water to reduce waste?

Posted by u/kellun133 • 0 points • 10 comments

basically the title. It's the #1 argument I hear against its environmental impact but I'm not sure exactly how it works or how water is exactly used in the process. If water is used as a source of energy (I'm thinking like how those energy generating dams work,) couldn't we just use dirty or non-potable water for the process? I hate AI even though I use it sometimes, so I've been looking for more sustainable programs but nothing really comes up. :/

Comments

u/LCplGunny • 1 points • 2025-03-11 18:22 UTC

Who the fuck is using an open loop for computer cooling? Most computer cooling is done with a closed loop. This means you don't get measurable losses on water volume. You're not refilling it, it's the same water running in a circle! It's also generally not that much water, tho I'd assume it would be more for an AI, than a home PC. It still shouldn't be THAT much water.

u/Jonatan83 • 2 points • 2025-03-11 18:30 UTC

With closed loop water cooling you're not really getting rid of any heat, you're just moving it to a heat sink. Big server halls use water to transport heat away from the building itself, not necessarily to cool individual computers.

u/LCplGunny • 2 points • 2025-03-11 18:37 UTC

Not trying to be mean or rude... But wtf does a heat sink do, if not remove heat from the loop?

As for the none closed loop thing... There has to be a better way to do it... Like I literally can't think of a bigger waste of water than a single run cooling system.

u/Jonatan83 • 1 points • 2025-03-11 19:28 UTC

A heat sink (or actually a radiator) is just a way to get rid of heat through air cooling. The heat would still be in the building, just not on the chip that is being cooled. Eventually the building would get so hot that things start failing.

> As for the none closed loop thing... There has to be a better way to do it... Like I literally can't think of a bigger waste of water than a single run cooling system.

It really depends. Not all places are short on water. And it's not like the water is destroyed. Additionally, in some places the waste heat is used to generate power or heat other buildings.

u/LCplGunny • 1 points • 2025-03-11 20:03 UTC

I could get behind that, that gives me a great idea to heat my future house... I'm a USAican, so I might get to buy a house during the next housing collapse if I have enough saved.

u/shootYrTv • 2 points • 2025-03-11 17:57 UTC

To the extent possible, waste water IS used to cool down these massive server farms. It still needs to be treated because you can't really have raw sewage running through coolant pipes.

The issue is that this massive amount of water is evaporated into the air and eventually makes its way back to the sea, further draining a resource that California is specifically already limited on.

u/azuth89 • 1 points • 2025-03-11 17:55 UTC

Water is for cooling and using dirty water will result in mucked up, clogged and damaged cooling systems on top of being less efficient.

I have to admit I've only heard that as a criticism in isolated circumstances like building them in a desert where water is already scarce. Generally the biggest criticism I hear is the amount of power they demand.

u/Chemical_Golf_2958 • 1 points • 2025-03-11 17:55 UTC

AI doesn't use water to generate energy like hydropower dams do. Instead, water is primarily used in data centers for cooling purposes. When AI systems operate, they generate a lot of heat. To keep the equipment from overheating, water is used to transfer this heat away. The water absorbs the heat and is then cooled down, often using chillers or cooling towers, before being recirculated.

Importantly, the process of transferring heat to the water doesn't make the water dirty. Therefore, the concept of using dirty or non-potable water to reduce waste doesn't quite apply here, as the water remains relatively clean, just heated.

That being said, there are ongoing efforts to reduce the environmental impact of AI and data centers. These include improving energy efficiency, using renewable energy sources, and developing advanced cooling techniques that use less water.

\#notsponsored

u/anti-beep • 2 points • 2025-03-11 17:53 UTC

The water is used for cooling, and dirty water will quickly cause impurities to build up, this requiring much more maintenance. You have to use clean water, and if you wanted to treat dirty water for use in cooling systems then that'd just mean spending more energy.

u/wt_anonymous • 3 points • 2025-03-11 17:48 UTC

You need water to cool things down. If all that dirt and sludge gets into whatever system is used to cool it down, that's bad.